ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · W1 · WORKING SCIENTIFICALLY

Tools and steps

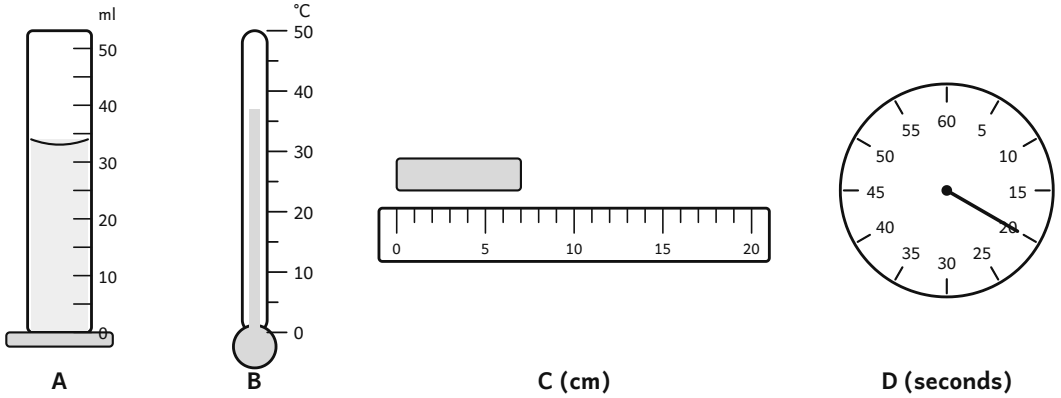
★☆☆
Grade 4

Learning goals

- Choose equipment and units.
- Order the steps of an investigation.

1 Read each instrument. Include the unit.

[4]



A B C (cm) D (seconds)

A	B	C	D
34 ml	37 °C	7 cm	20 s

2 Match the measurement to the tool.

[5]

Draw a line to match each pair, or write the letter.

how hot the water is ●	→ E	● **A** ruler
how long a race takes ●	→ D	● **B** measuring jug
how long a leaf is ●	→ A	● **C** forcemeter
how much juice is in a glass ●	→ B	● **D** stopwatch
how hard you pull a rope ●	→ C	● **E** thermometer

3 **Put the steps of an investigation in order.** [4]

- 5 record the results
- 1 ask a question
- 6 write a conclusion
- 4 do the test and measure
- 2 make a prediction
- 3 plan a fair test

4 **Bar chart or line graph?** [3]

Results	Best graph
favourite fruits of our class	bar chart
temperature of the classroom every hour	line graph
how far four kinds of balls bounce	bar chart

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · W1 · WORKING SCIENTIFICALLY

Is it fair?

★★☆
Grade 4

Learning goals

- Spot unfair tests.
- Write predictions.
- Find patterns.

1 **Sami wants to find out which paper towel soaks up the most water. He uses a big piece of towel A and a small piece of towel B, and dips them for different times. Is this fair? Explain how to make it fair.** [3]

No: he changed the size and the time as well as the type of towel. Use pieces of the same size, dip them for the same time in the same amount of water, and change only the type of towel.

2 **Write a prediction with a reason: "Will an ice cube melt faster in the sun or in the shade?"** [2]

I think it will melt faster in the sun because the sun gives it more heat.

3 **Look at the results. What is the pattern?** [2]

Amount of water each day (ml)	0	25	50	100
Height of bean plant after 2 weeks (cm)	2	9	15	16

The more water the plant gets, the taller it grows, but above 50 ml it hardly grows any taller.

4 **Three children timed the same race: 12 s, 13 s and 25 s. Which result is odd? What should they do?** [2]

25 s is odd (maybe the stopwatch was started late); they should time the race again.

ANSWER KEY

INVESTIGATE · W1 · WORKING SCIENTIFICALLY

Does a higher drop make a higher bounce?

★★☆ Grade 4

Learning goals

- Plan, measure and repeat.
- Draw a line graph.
- Conclude and evaluate.

Class 4A dropped a tennis ball from different heights onto the same floor and measured how high it bounced. They did each drop three times.

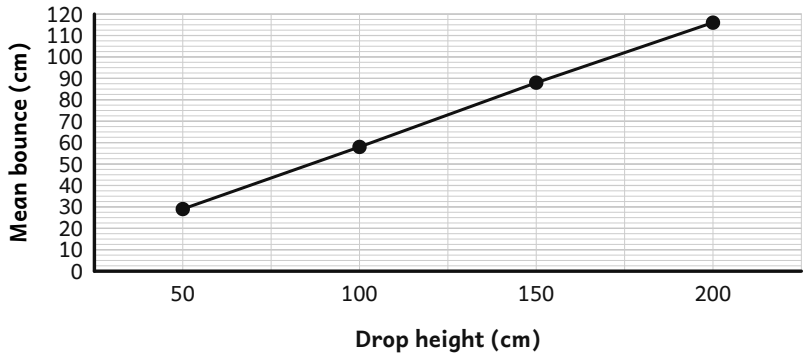
1 Complete the plan. [4]

Question	Does a higher drop make a higher bounce?
Change	height of the drop
Measure	height of the bounce (cm)
Keep the same	the same ball, floor, and way of dropping (no throwing)
Prediction	the higher the drop, the higher the bounce

2 Complete the means. [4]

Drop height (cm)	Bounce 1	Bounce 2	Bounce 3	Mean bounce (cm)
50	28	30	29	29
100	57	60	57	58
150	86	88	90	88
200	115	118	115	116

3 Draw a line graph of the means. [3]



4 **Write a conclusion. Did the results support the prediction?** [2]

Yes: the higher the drop, the higher the bounce. The bounce was always a bit more than half the drop height.

ANSWER KEY

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LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · W1 · WORKING SCIENTIFICALLY

Dr Firdausi Qadri: science that saves lives

★★★
Grade 4

Learning goals

- Learn how a real scientist uses evidence.
- Read data from a trial.
- Think about how science helps people.

Science | Health | Maths | Citizenship

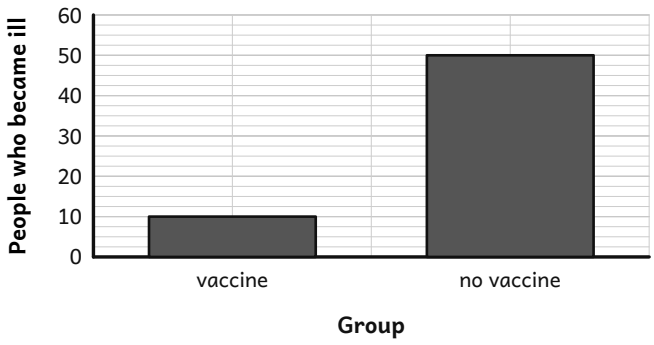
Cholera and **typhoid** are diseases spread by dirty water. They can make children very ill. **Dr Firdausi Qadri**, a scientist at **icddr,b** in Dhaka, has spent her life finding ways to protect people from them.

Scientists cannot just guess that a vaccine works; they need **evidence**. So Dr Qadri and her team ran large **trials** with thousands of volunteers in Dhaka. Some people got the vaccine and others did not, and the team counted how many in each group became ill. The evidence helped make cheap vaccines available to millions of children. In 2021 she won the Ramon Magsaysay Award, one of Asia's top prizes, for her work.

Based on information about Dr Firdausi Qadri and icddr,b. Numbers below are example data.

1 **A vaccine trial (example data): 1000 people in each group.** [3]

Group	Got the vaccine	Did not get the vaccine
People who became ill	10	50



a) Draw the bar chart.

b) How many times more people became ill without the vaccine?

50 ÷ 10 = 5 times more

2 **Why did the trial need two groups?** [2]

To compare: the group without the vaccine shows what would happen anyway, so we can see if the vaccine makes a difference.

3 **Why was it important to test thousands of people, not just ten?** [2]

With many people, the results are more reliable and not just luck.

4 **Besides vaccines, how can people protect themselves from diseases spread by dirty water?** [2]

Drink boiled or treated water; wash hands with soap; use clean toilets.

ANSWER KEY

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REVIEW AND CHALLENGE · W1 · WORKING SCIENTIFICALLY

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

M	O	L	M	R	D	F	A	I	R
T	E	V	E	O	H	C	S	E	M
N	V	P	A	B	V	F	P	R	F
S	I	A	S	S	T	E	R	E	H
J	D	T	U	E	E	P	E	S	G
U	E	T	R	R	S	G	D	U	R
J	N	E	E	V	T	O	I	L	A
L	C	R	E	E	M	J	C	T	P
B	E	N	A	O	W	D	T	O	H
D	H	O	K	S	L	H	F	F	B

evidence fair graph measure observe pattern predict result test

2 In a fair test you change... [1]

○ everything
● **only one thing**
○ two things
○ nothing

3 Which tool measures temperature? [1]

○ ruler
○ stopwatch
● **thermometer**
○ forcemeter

4 Plan your own investigation: "Which ball bounces highest?" Write what you will change, measure and keep the same. [3]

Change	the type of ball
Measure	the height of the bounce (cm)
Keep the same	drop height, floor, how it is dropped, who measures

5 Challenge. Why do scientists share their results with other scientists? [2]

So others can check the results by repeating the tests, find mistakes, and build on the ideas to make new discoveries.